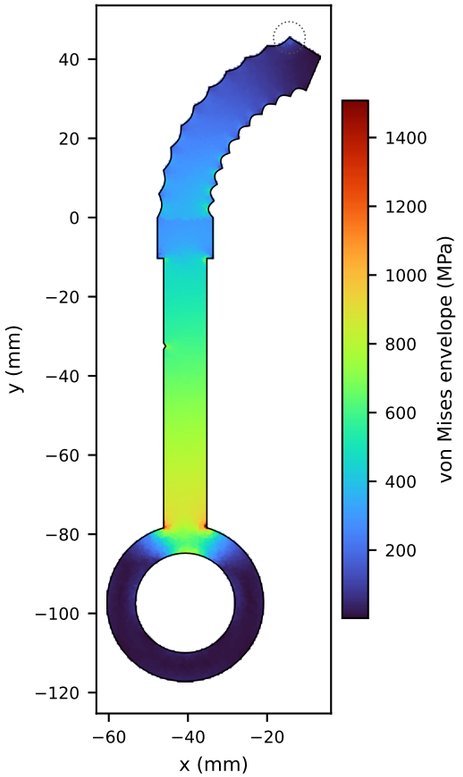
Wave 2 — bench10k (MC 10k benchmark) | Fitness 6.797 N/g | Est. weight 44.9 g

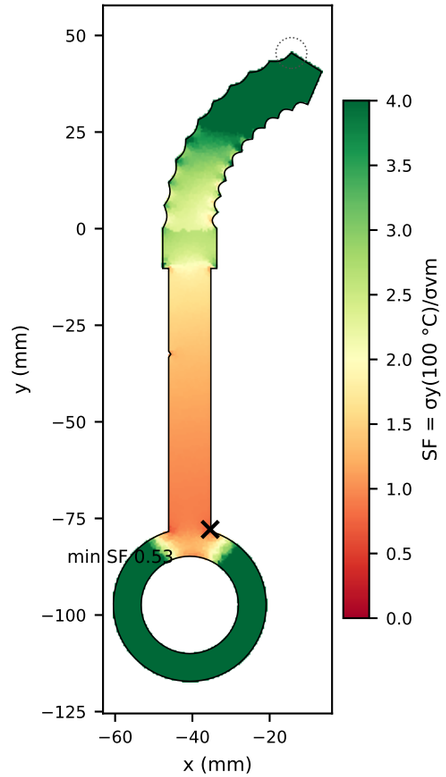
FEM-UNCALIBRATED: offline Kirchhoff plate (numerical DKT + CST, 6156 tris) — not yet calibrated to MC / repo FEM. Envelope of hunter-lateral 260 N, twist 1.26 N·m, tip-pry 260 N, ring-pull 1500 N.

FEM-UNCALIBRATED **QUARANTINED: blade < 3.375 in (all three)** **hard geometric gates: red-team Ok** **floors 180/260/1500 N on plate: FAIL**

(1) von Mises — worst load case per element



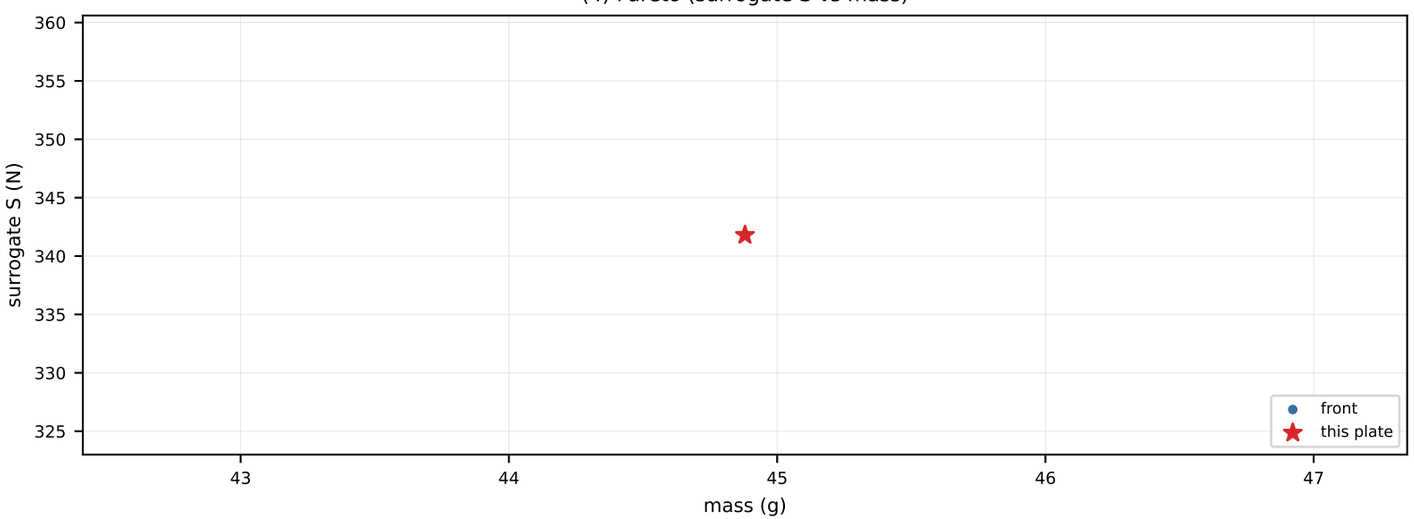
(2) Safety factor — worst case @100 °C



(3) Metrics

Metric	Value	Gate / note
Blade tip→heel	2.276 in (57.8 mm)	**≥3.375 in: FAIL**
Blade horizontal reach	1.620 in (41.1 mm)	**≥3.375 in: FAIL**
Blade edge arc (outer)	2.737 in (69.5 mm)	**≥3.375 in: FAIL**
Grip (ring-neck→guard)	68.1 mm	60–80 mm
Ring ID	25.19 mm	25.4±0.5 mm
Stock / material	4.4 mm / Ti-6Al-4V	red-team: Ok
Est. weight	44.88 g	geom check 44.88 g
Fitness	6.797 N/g	MC p05 strength-per-gram
Beam surrogate S	342 N	repo plate (fem.rs): n/a
Plate FEM @ -10 °C	lat 157 N | ring 8404 N	SF hiker 0.87 / hunter 0.60 / ring 5.60
Plate FEM @ +20 °C	lat 152 N | ring 8135 N	SF hiker 0.84 / hunter 0.58 / ring 5.42
Plate FEM @ +100 °C	lat 138 N | ring 7420 N	SF hiker 0.77 / hunter 0.53 / ring 4.95
Twist / pry SF @100 °C	twist 11.19 | pry 1.53	twist 1.26 N·m, pry 260 N

(4) Pareto (surrogate S vs mass)



Floors on offline plate, worst of -10/20/100 °C: hiker FAIL · hunter FAIL · ring PASS
Temp: Ti yield linear derate (-0.11 %/°C from 20 °C); maps use 100 °C yield 803 MPa.
Serration knock-down on blade (h/h_eff=1.17, from history genes — DEFAULT 1.0/1.0 mm). Load patch r≤4 mm exclu